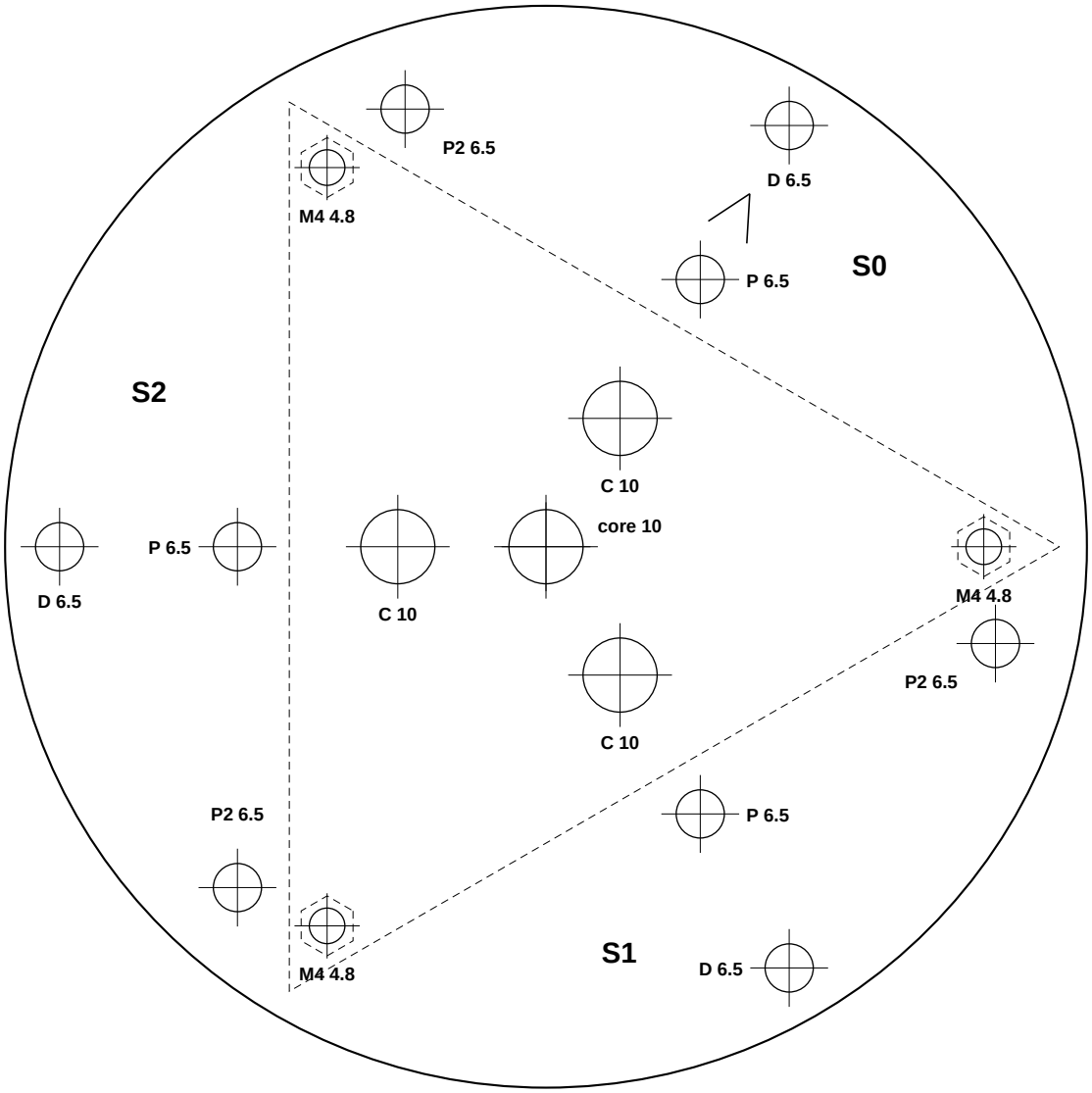
# Shield deck drill template - 146 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP (electronics side). Tape to the board, centre-punch each crosshair, then drill.




Solid circle = cut line (146 mm). Dashed triangle = where the 120 mm back panels stand, copper side out.

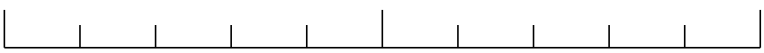
D 6.5 mm = rubber duck / monopole (31 mm in front of face) P / P2 6.5 mm = patch (7 mm in front of face; P2 14 mm from the face end)

C 10 mm = dipole coax pass-through (inside core) M4 4.8 mm = bolts to the plate, nut flush in each core corner

Core 10 mm at the centre = antenna coax into the core; open it up with a hole saw (centre mark) if needed.

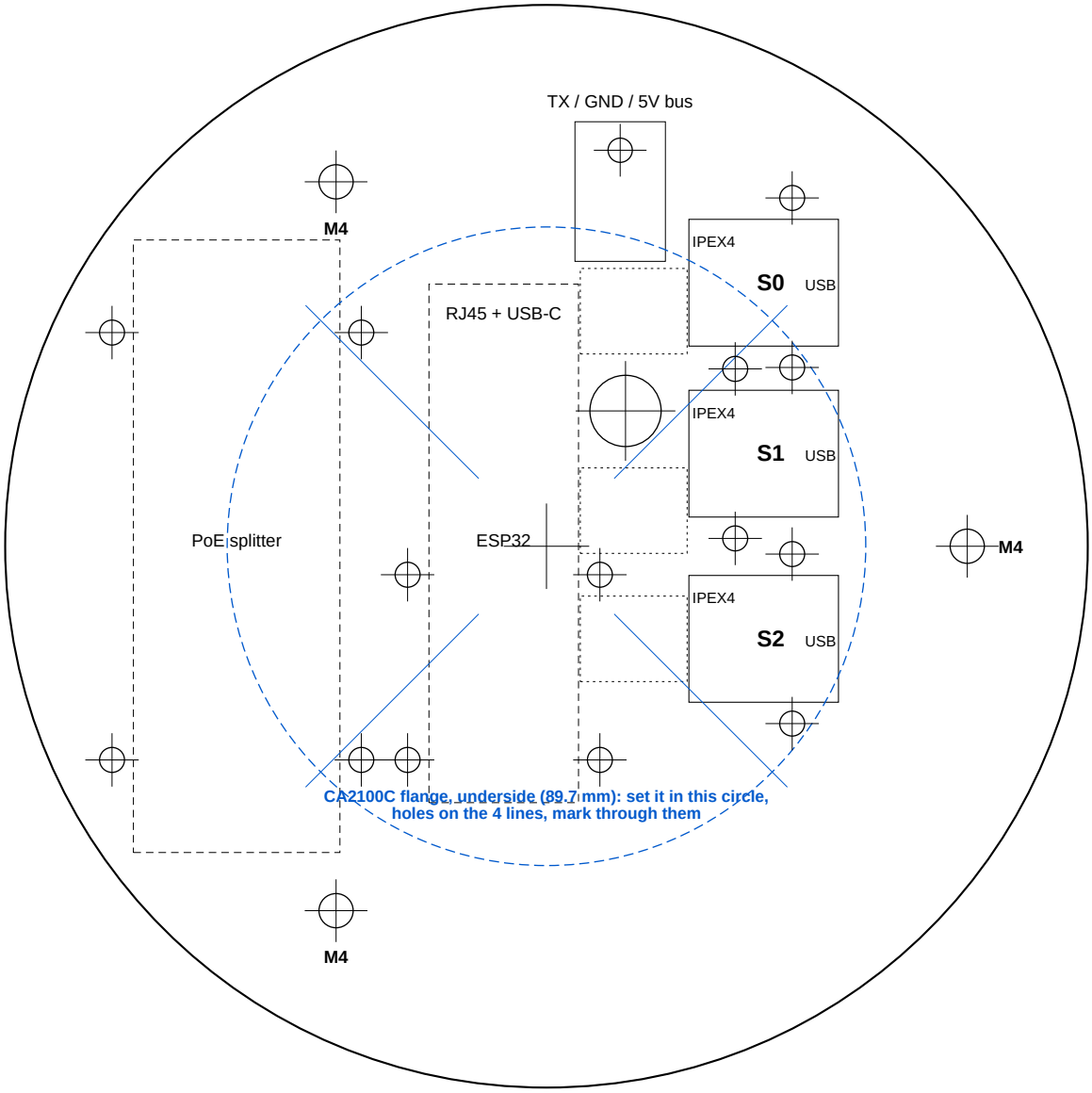
Arrow points to Sector 0. Holes are symmetric, so the template also works flipped (labels read mirrored).

Drawn the same way round as the plate (page 2): plate 12 o'clock up, S0 arrow 30 deg clockwise of it.

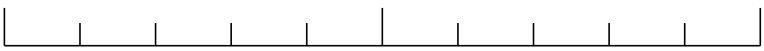


# Electronics plate template - 152 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP (parts side). Gateway left, scanners right, one pigtail hole and the JST pairs between.

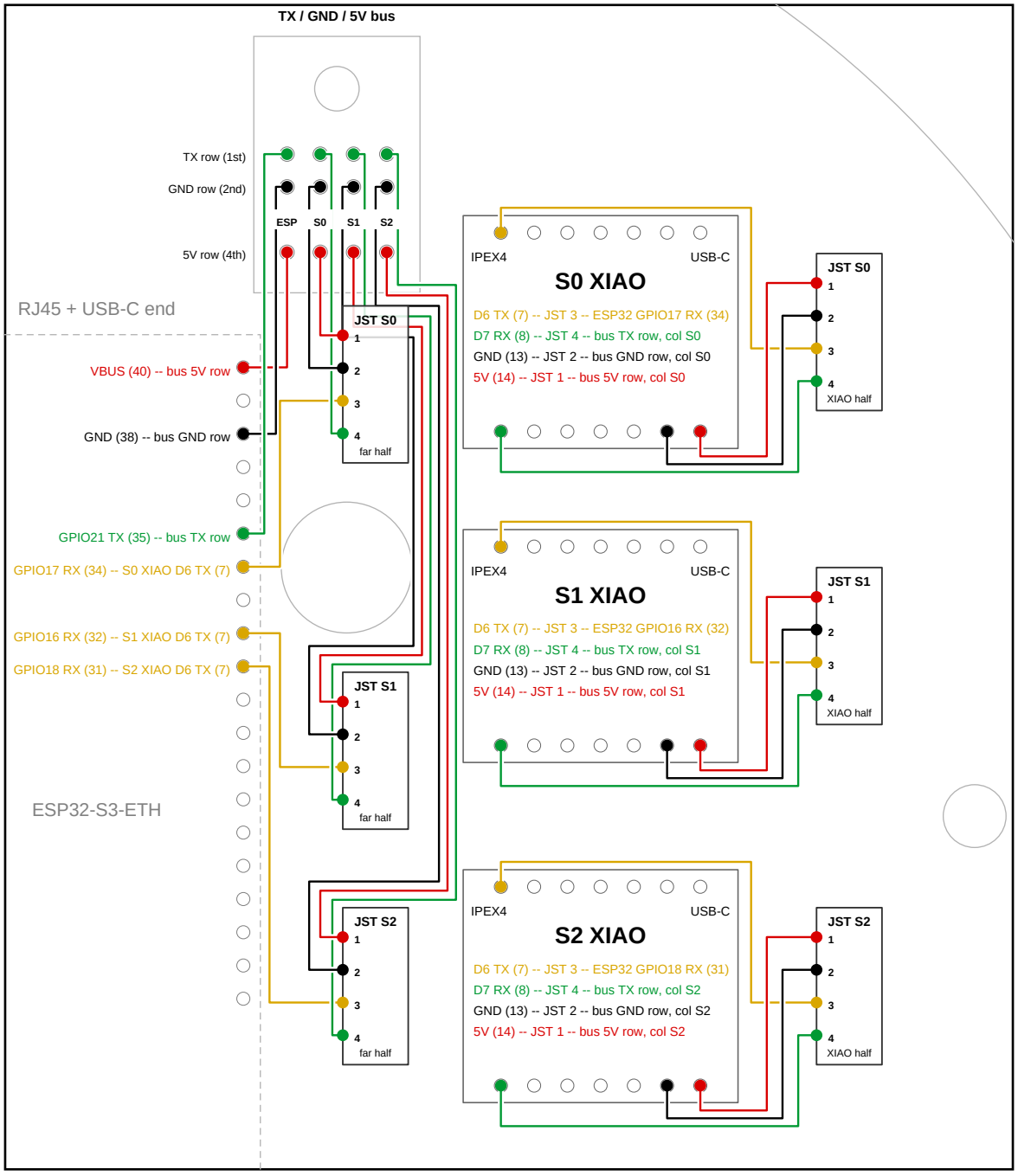



10 mm: all three pigtails (MHF4 ends threaded up from below), level with S1's IPEX4 connector - assumed 6 mm above the XIAO centre line: check on the board. 3.4 mm (M3): bus board.
3 x 4.8 mm (M4, labelled): bolts to the deck, plate 12 o'clock 30 deg counter-clockwise of the deck's S0 arrow, ~25 mm up.
Small holes: 3.5 mm zip-tie pairs: one tie across each XIAO (USB-C toward the rim), two each across splitter and ESP32. Coax strain relief: grommet in the pigtail hole, or a dab of RTV once the coax are placed.
Dotted: mated JST pairs (unplug to reflash); far side D6 to its ESP32 RX pin, 3 wires to the bus.
Dashed: ESP32-S3-ETH 72.8 x 21 mm; splitter from a photo - lay it on the printout, tie holes 3 mm outside its edges.



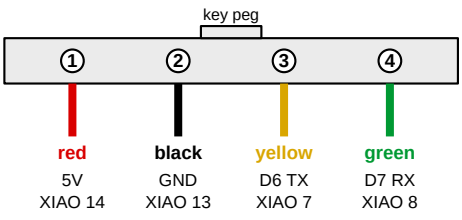
# Electronics plate wiring - same layout as the drill template

Top view of the radio side, 2x (not for drilling). Pin numbers in brackets. Pad positions approximate.
JST halves drawn apart: XIAO half right of each radio, far half in the strip; same name = they plug together.




D6 -> its own ESP RX (the pin is the sector): S0 GPIO17 (34), S1 GPIO16 (32), S2 GPIO18 (31).
ESP TX GPIO21 (35) -> bus TX row -> every D7. Feed: TX (35) -> TX row, VBUS (40) -> 5V row, GND (38) -> GND row.
Bus: hole end up, rows bridged, 3rd row spare. ESP32 RJ45 side up, RJ45 / USB-C end north: VBUS (40) top right pin.

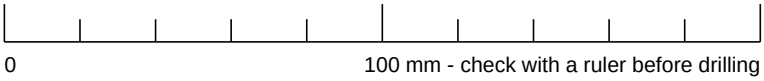
**JST head (either half), crimp side, key peg up:**

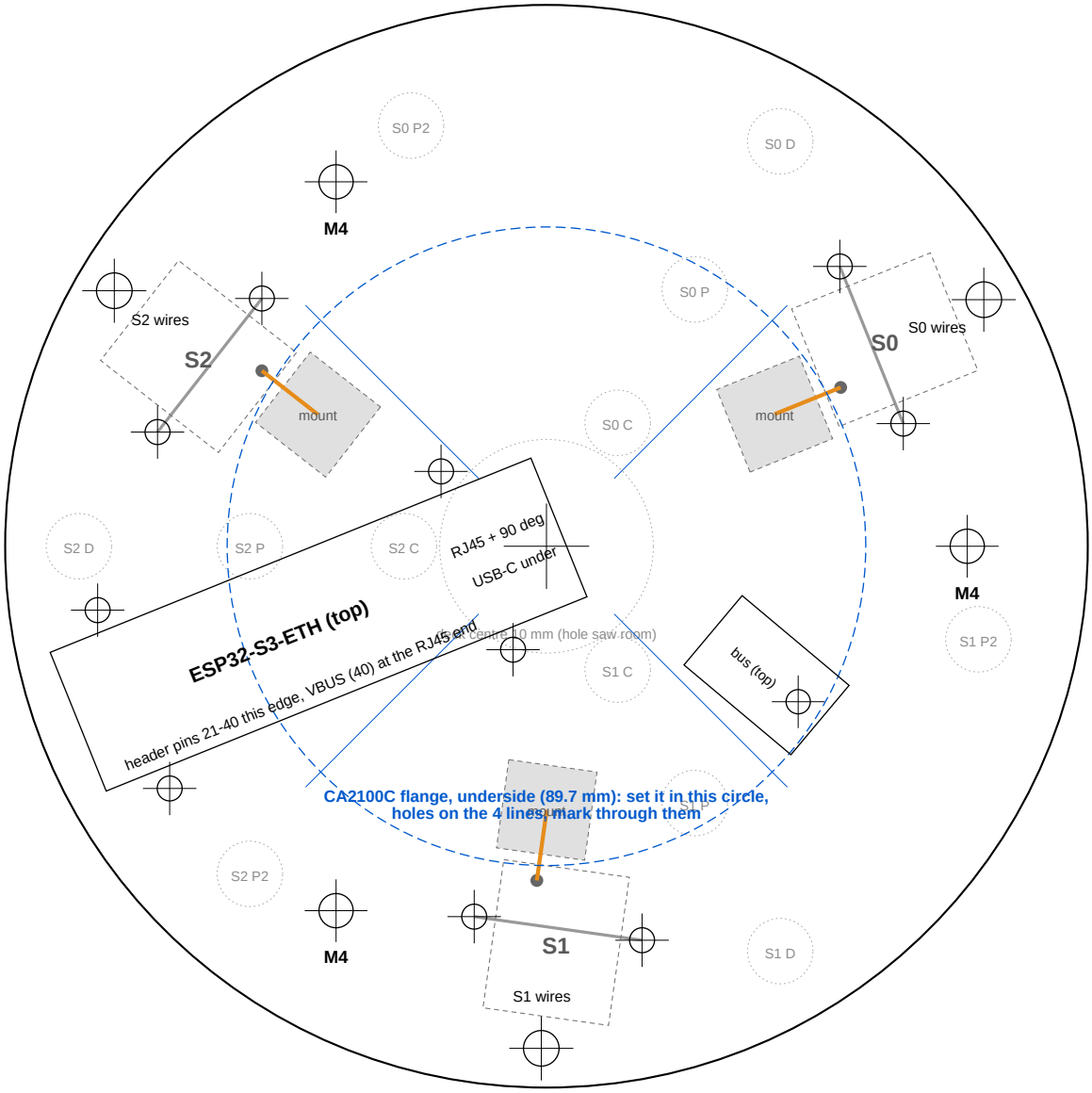



Pins left to right: 1 red 5V, 2 black GND, 3 yellow D6 TX, 4 green D7 RX.
Male and female halves carry the same colours in the same order on the crimp side, so the XIAO tail and the far tail match colour for colour through the joint.
XIAO tail: red -> 5V (14), black -> GND (13), yellow -> D6 (7), green -> D7 (8).
Far tail: red -> bus 5V row, black -> bus GND row, green -> bus TX row, yellow -> that sector's ESP RX pin (labels on the drawing). Female half on the far tail keeps 5V covered.

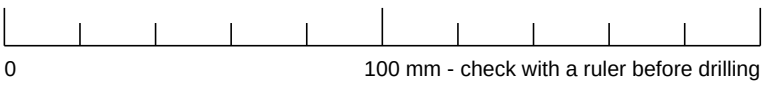
# ALT electronics plate (split) - 152 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP. Radios on the UNDERSIDE (grey, seen through the plate); ESP32 + bus on top. No pigtail hole.



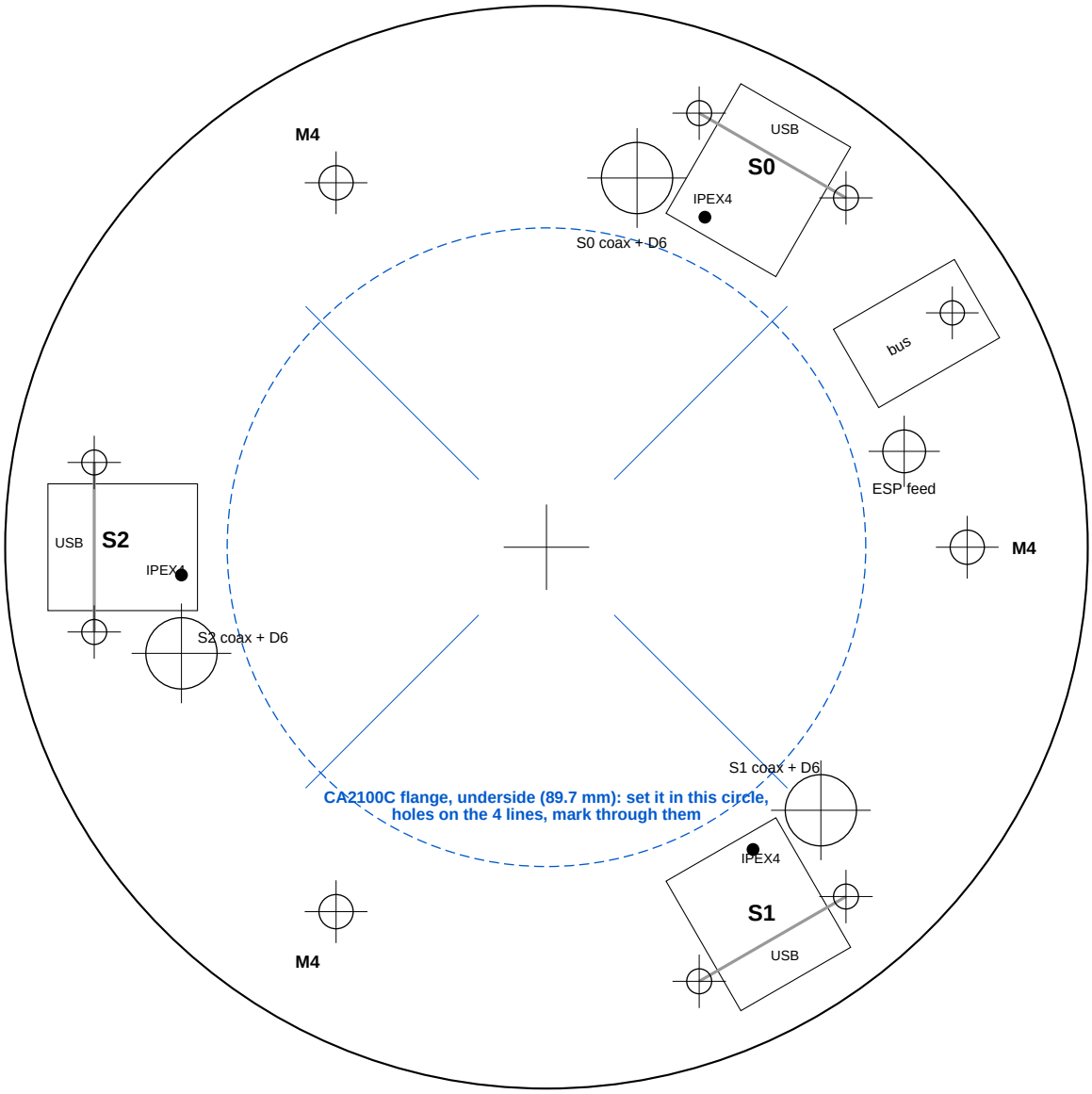




Drill: 3 x 4.8 mm (M4) bolts to the deck, same as page 2; 3.4 mm (M3) bus board; 3 x 5 mm harness holes (chamfered): each XIAO's 4-wire tail up to its JST pair on top; 3.5 mm pairs: one tie across each XIAO, two across the ESP32. Underside (grey): XIAOs on VHB + a zip tie (grey bar), IPEX4 dot -> orange flat coax run -> HS-100 tie mount, which clamps it; bend only after the mount, down to its deck jack. Dotted circles: deck jacks below, not drilled. Top: ESP32 parallel to S0, RJ45 / USB-C end toward 2 o'clock. A 90 deg adapter turns the RJ45 up; patch + USB-C lead run through the 4" box's 1" hole to the PoE splitter. USB-C is under the board: slim plug. Spacers 25-30 mm. S1 / S2 are S0 turned 120 deg about the centre. Underside keeps clear of the deck's centre hole (30 mm, assumed).



# FLANGE electronics plate - 152 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP. XIAOs + bus on top, flange underneath; PoE splitter + ESP32 inside the core.

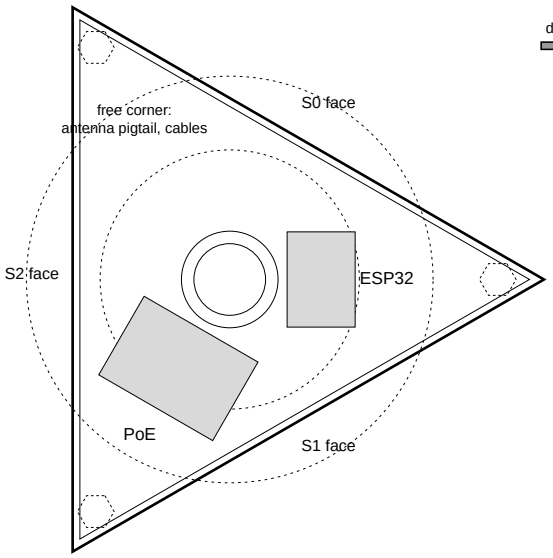



Drill: 3 x 4.8 mm (M4) bolts to the deck, same as page 2; 3.4 mm (M3) bus board; 3 x 10 mm pigtail holes; 6 mm ESP feed hole (TX, GND, VBUS up from the core to the bus); 3.5 mm pairs: one tie across each XIAO. Flange underneath (blue): set it in the circle, holes on the lines, mark through. Keep top-side parts out of the circle: flange bolt heads. The PoE cable passes the plate rim (the pipe is capped: see the stack views). Each XIAO sits over its own sector: pigtail up its hole to IPEX4, its D6 wire back down the same hole to its ESP32 RX pin; 5V, GND and D7 over the top to the bus. IPEX4 (dot) is on the right seen from the top with USB-C toward you, so each pigtail hole is on that side. USB-C toward the rim, clear of the M4 bolts.

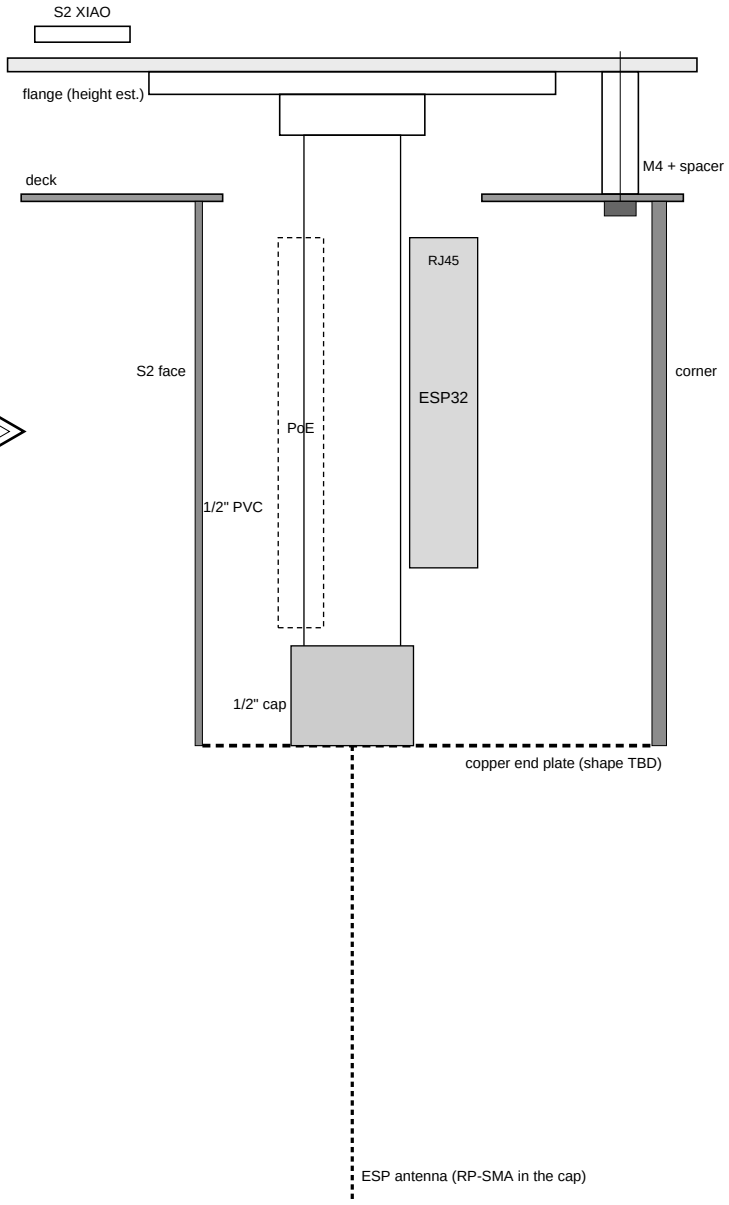
# Flange stack - top and side views - scale 0.6 (NOT a drilling template)

Indoor way up: electronics plate on top, core and antennas below. Same plate frame as page 5 (12 o'clock up).

**TOP: inside the core, looking down**




**SIDE: cut through 3 - 9 o'clock**




dotted: flange (above the deck)
and the 2-1/4" deck hole;
dotted hexes: M4 nuts under the deck

Pipe: 1/2" PVC (21.3 mm OD) from the flange (male adapter) through the 2-1/4" deck hole, ~145 mm long (check the flange's thread depth), capped flush with the core's far end. The cap holds the ESP antenna's RP-SMA bulkhead; a U.FL pigtail runs from the ESP32 through a hole in the pipe wall down to it. The cap also anchors the copper end plate (shape TBD). Splitter and ESP32 zip-tied to the pipe facing the 3 and 7 o'clock corners (widest room), RJ45 / USB-C end up; the 11 o'clock corner is left for the antenna pigtail and cables. PoE cable: down past the plate rim, through the deck hole. Side: dashed box = PoE splitter, behind the cut. XIAOs, pigtail holes, bus board: as page 5. Estimates: flange height, splitter size, board depths, antenna length.

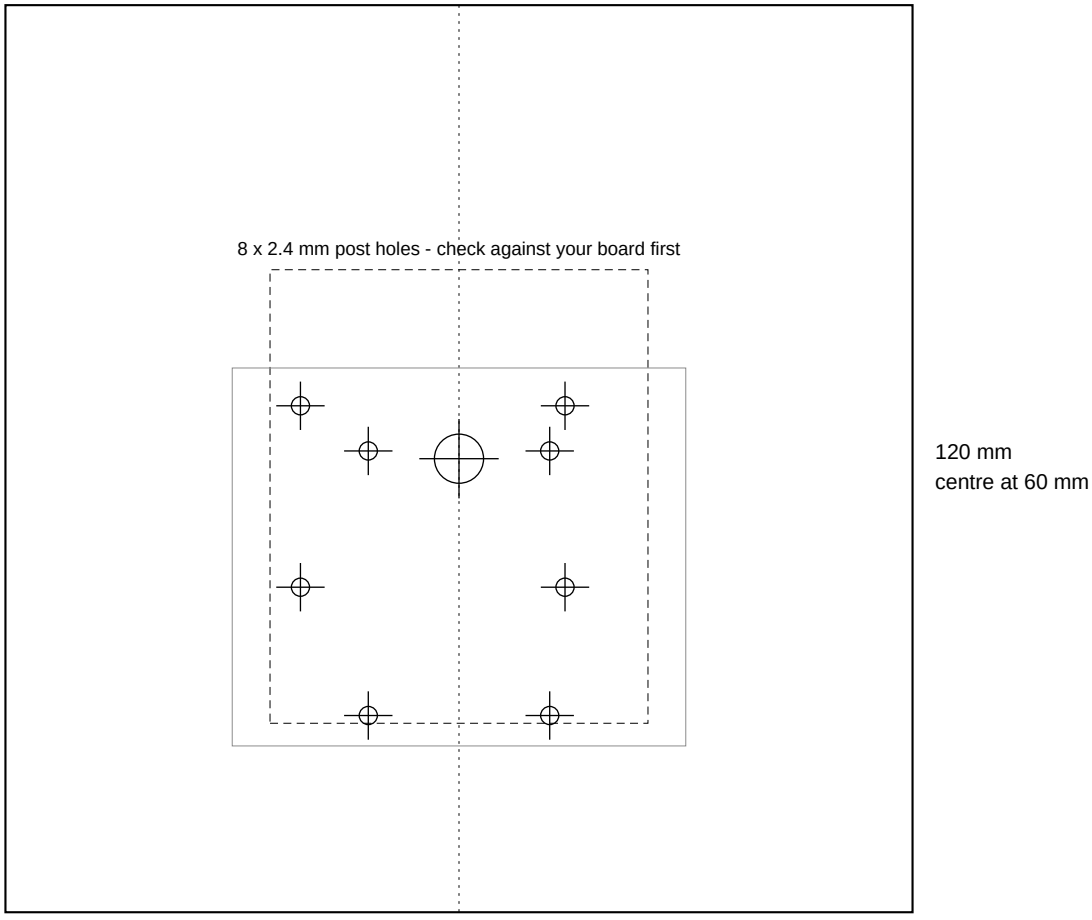
50 mm at drawing scale

# Back panel (reflector face) template - 120 x 120 mm - PRINT AT 100%

Make 3. Viewed from the FRONT (copper side, facing the antenna). Top edge goes against the shield deck.

0 100 mm - check with a ruler before drilling

**TOP - against shield deck**




A (dashed): bar vertical, notch right, board 35-95 mm down - stalk reaches the P2 deck jack

C (grey): bar horizontal, notch up, board 48-98 mm down

Centre 6.5 mm hole: DIY dipole RP-SMA only

Rubber duck / monopole / dipole sit 31 mm in front of this face (on the D hole).

Panel board 50 x 60 mm: 7 mm off this face on nylon standoffs. A = stalk on the P2 deck jack; C = coax may need extending.

Corners meet at 120 deg to form the triangle; solder or tape the seams. Holes under 10 mm don't affect the reflector.

0 100 mm - check with a ruler before drilling